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Introduction to Generative AI

Diffusion Models and Image Generation

Outline

- Module Learning Objectives
- Foundational Models of Image Generation
- GANs in Practice
- Emergence of Diffusion Models
- Large Vision Models

Module Learning Objectives

- Adapt GAN, VAE, spatio-temporal GAN to generate image and video
- Develop and fine-tune models via pre-trained models, examples will be provided from Hugging Face and PyTorch models
- Measure and assess performance by pixel-wise loss functions, inception score (IS) and Inception distance (FID) scores

Foundational Image Generation

- Generative Adversarial Networks
  - GANs use a two-player minimax game btw a generator G and a discriminator D
  - $\min_G \max_D V(D,G) = \min_G \max_D \mathbf{E}_{\boldsymbol{x}\sim \mathrm{P}_d}[\log D(\boldsymbol{x})] + \mathbf{E}_{\boldsymbol{z}\sim \mathrm{P}_z}[\log(1 - D(G(\boldsymbol{z})))]$
  - **Challenges** are mode collapse and training instability
- Variational Autoencoders
  - Model latent space z through an encoder $\mathrm{q}(z \mid x)$ and decoder $\mathrm{p}(x \mid z)$
  - **Loss Function** is evidence lower bound (ELBO)
  - $\mathrm{Loss} = -\mathrm{KL}(\mathrm{q}(z \mid x) \parallel \mathrm{p}(z)) + \mathbf{E}_{\mathrm{q}(z|x)}[\log \mathrm{p}(x \mid z)]$
  - **Limitations** lower image fidelity compared to GANs
- `scipy.stats.entropy(P,Q)`
  - Empirically computes the KL divergence between $\mathbf{P}$ and $\mathbf{Q}$ vectors

GANs in Practice: StyleGAN to StyleGAN3

- StyleGAN can control and refine the visual attributes of generated images
  - Such as facial expressions, lighting, background, in various levels of abstraction
- StyleGAN (2019)
  - Latent Space and **Style Mixing** can take a sample latent vector z and map it into an intermediate latent space w via a learned mapping network
  - Certain layers of the G control high-level attributes (like the shape of the face), while other layers control finer details (like skin texture or lighting)
  - **Progressive Growing** can start G by creating low-resolution images and gradually increases the resolution by adding layers during training
- Applications include high-quality facial images that are indistinguishable from real photographs, generating synthetic portraits, art, and even video game characters

StyleGAN2 (2020)

- **Elimination of Water Droplet Artifacts** certain image regions would exhibit unnatural and repetitive patterns and solved via normalization and scaling techniques
  - Replaced instance normalization with direct adjustment, weight demodulation
- **Improved Fidelity** by better decoupling high-level features (such as the overall structure of a face) from fine-grained details (such as wrinkles or textures)
- Applications include high-quality, photorealistic image generation even adopted for generating faces for avatars, digital marketing materials, and automated content creation tools

StyleGAN3 (2021)

- **Aliasing Artifacts** appear when attempting to generate smoothly transforming images, such as in videos or animations
- **Continuous Transformations and Improved Sampling** ensures that small changes in the latent space result in smooth changes in the generated images
- **Elimination of Aliasing** visual distortions that occur when high-frequency signals are sampled inadequately solved with better spatial consistency
- Applications include video generation, where temporal consistency is essential

Emergence of Diffusion Models

- Diffusion models generate images by gradually denoising random noise, reversing a Markovian forward process that corrupts data with noise
- **Forward Process** sequentially adds Gaussian noise to an image x_0 over T steps
  - Process is $\mathrm{q}(x_t \mid x_{t-1}) = \mathbf{N}(x_t; \sqrt{\alpha_t} x_{t-1}, (1-\alpha_t)\mathbf{I})$
  - $\sqrt{\alpha_t} x_{t-1}$ is the mean and $(1-\alpha_t)\mathbf{I}$ is the variance in multivariate normal
- **Reverse Process** learns to denoise starting from $x_T \sim \mathbf{N}(0, \mathbf{I})$ back to x_0
  - Minimize variance in each step $\mathrm{p}_\theta(x_{t-1} \mid x_t)$ via denoising score matching
  - Advantage is high fidelity and diversity in generated images
  - Use a deep neural network optimizing a variational lower bound, ELBO, on log-likelihood

Score Matching in Diffusion Models

- Learn a model for the score function $\nabla_x \log p(x)$ rather than modeling $p(x)$ directly
- Predict the gradient $s_\theta(x) \approx \nabla_x \log p(x)$
  - where $s_\theta(x)$ is a NN approximates the true score function
  - Avoids computing the normalization constant of $p(x)$, which is often intractable
- Once the score function learned, sample using Langevin dynamics

  $x_{t-1} = x_t + \eta \cdot s_\theta(x_t) + \sqrt{2\eta} \cdot z$ and $z \sim \mathbf{N}(0, \mathbf{I})$

- This process **gradually denoises** an image from pure noise

Langevin Dynamics

- Introduced by a French physicist Paul Langevin in 1908
- A mathematical framework to describe the motion of particles in a fluid under the influence of **random thermal fluctuations** and **friction**
- Statistical mechanics, particularly modeling **Brownian motion** using a stochastic differential equation

$$m\frac{dv}{dt} = -\gamma v + \sqrt{2D} \cdot \xi(t)$$

  - m is the particle's mass, γ is a friction coefficient, D is the diffusion constant, $\xi(t)$ is Gaussian white noise
- Adapted for sampling from probability distributions, particularly in Markov Chain Monte Carlo (MCMC) and score-based generative models

Training Objective

- Instead of directly modeling μ_θ, predict the noise ϵ using a NN ϵ_θ, then training loss,

  $$L_{\text{simple}}(\theta) = \mathbf{E}_{t,x_0,\epsilon}\left[\| \epsilon - \epsilon_\theta\left(\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t\right) \|^2\right]$$

  - Equivalent to denoising score matching, the model learns to estimate the noise
- Connection to Langevin Dynamics and sampling (image generation)
- Given pure noise $x_T \sim \mathbf{N}(0, \mathbf{I})$, the image is generated using

  $$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(xt - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t)\right) + \sigma_t z$$

  - $x_T \sim \mathbf{N}(0, \mathbf{I})$ adds stochasticity to the process
- The model effectively learns the score function of the data distribution $\nabla_x \log p(x)$

U-Net for Diffusion Models

- U-Net is a convolutional neural network designed for image-to-image tasks
- Originally introduced for medical image segmentation, now widely used in diffusion models as the denoising network that predicts noise at each step
- **Encoder** downsampling path
  - Series of convolutional layers followed by max pooling
  - Captures global features while reducing spatial resolution
- **Bottleneck** latent representation
  - The deepest layer of the network
  - Extracts high-level features from the noisy input
- **Decoder** upsampling path
  - Uses transpose convolutions to restore image resolution
  - Includes skip connections from the encoder to preserve fine details

Large Vision Models and SAM

- **Segment Anything Model** (SAM) is trained on massive datasets for adaptable segmentation
  - Architecture is encoder-decoder with dense feature representation
  - Allows precise, generalizable segmentation across diverse domains
  - Performance maintains segmentation accuracy with efficient fine-tuning on new image domains
- Trained on SA-1B with over 1 billion masks on 11 million images
  - Extensive data diversity enables SAM to generalize with minimal to no fine-tuning
- SAM architecture allows it to perform zero-shot segmentation effectively

High-Quality Image Generators

- **Stable Diffusion** employs a latent diffusion model for efficient, high-resolution image generation
- **Mathematical Model** is diffusion in latent space, optimized using conditional probability $\mathrm{p}_\theta(x_{t-1} \mid x_t, c)$ for image condition c
- Future Trends
  - Fine-tuning diffusion models for domain specificity
  - Advancements in memory-efficient diffusion processes for larger datasets
  - Emerging focus on **zero-shot generalization** across diverse image domains

Inception Score

- A metric used to evaluate the quality and diversity of generated images
- **Classification Confidence** via a pre-trained Inception v3 deep CNN model
  - Computes a class y probability distribution $\mathrm{p}(y|x)$ for each generated image x
  - A low entropy (or high confidence) would be achieved with high quality images
- **Diversity in Generated Images** measured from $\mathrm{p}(y)$ via marginal of $\mathrm{p}(y|x)$
  - A good spread of $\mathrm{p}(y)$ across many classes is expected
- **IS** is defined as the exponential of the Kullback-Leibler divergence between the conditional distribution $\mathrm{p}(y|x)$ and the marginal distribution $\mathrm{p}(y)$
  - $\mathrm{IS} = \exp\left(\mathbf{E}_{\mathbf{x}}[\mathrm{KL}(\mathrm{p}(y|x) \parallel \mathrm{p}(y))]\right)$
- A high KL divergence indicates that each image is strongly associated with a specific class and that the images are diverse

Fréchet Inception Distance

- Another popular metric for evaluating generative models, unlike the IS, FID explicitly measures the distance between real and generated image distributions
- **Feature Extraction** with Inception v3 both real and generated images are passed through a pre-trained Inception v3 model to extract feature representations
  - Transforms images into lower-dimensional vectors
- **Modeling Feature Distributions** where FID assumes multivariate Gaussian
  - Denote the mean and covariance as (μ_r, Σ_r) and (μ_g, Σ_g)
- **Calculating Fréchet Distance** or Wasserstein-2 distance defined as
  - $\mathrm{FID} = \left\|\mu_r - \mu_g\right\|^2 + \mathrm{Tr}\left(\Sigma_r + \Sigma_g - 2\left(\Sigma_r \Sigma_g\right)^{1/2}\right)$
  - Tr measures the similarity in the "spread" or shape of the two distributions
- Lower FID values indicate that the generated images are more similar to the real images in terms of both quality and diversity

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